

Benchmarking Quantum Computers via Verification Protocols

1 Benchmarking

Verification protocols are most commonly discussed as tools for certifying a single delegated computation. However, the information they produce — in particular, the failure rate of test rounds — is meaningful beyond this narrow use case. Under the sole assumption that the device behaves consistently from round to round (i.i.d. across rounds), this failure rate is a stable property of the device, independent of the specific computation being run. Extracting and interpreting this information is therefore an additional application of verification protocols, complementary to their primary role.

This observation was formalised by Frank et al. [?], who establish a rigorous link between verification and benchmarking and show that any verification protocol can be compiled into a certifiable benchmark for a class of computations. In particular, they propose characterising quantum devices by certifying 2D brickwork states of increasing dimensions (N, D) , yielding a device profile of the form shown in their Figure 4. In the present work we instantiate this approach concretely using the Veriphix verification framework: we run the FK verification protocol on simulated circuits of varying dimensions, and construct the resulting feasibility landscape explicitly as a function of the round budget R and security target ε . This benchmarking application is one of two utilities of verification protocols demonstrated in this paper, the other being noise estimation (Section ??).

1.1 From test rounds to a utility metric

When test rounds fail or pass, that carries information about the device’s noise. If the device behaves consistently throughout time (i.i.d. assumption), this failure rate is stable: it does not depend on which specific computation was run, nor on when the rounds were executed. Furthermore, since verification is blind by construction, the failure rate is the same for *all* computations sharing the same brickwork dimensions (N, D) — the device cannot distinguish between them.

The test-round failure rate therefore measures utility directly: it quantifies how well the device handles any computation of dimension (N, D) , in the worst case over all such computations. By asking whether the observed failure rate

is low enough to certify computation at dimension (N, D) with security ε in R rounds, one obtains a concrete, algorithm-independent performance metric. This is the benchmark.

1.2 Benchmarking protocol

Algorithm 1 Verification-based feasibility benchmark

Require:

- Set of circuit dimensions $\{(N_i, D_i)\}$ to probe
- Round budget R (total number of rounds the experimenter can afford)
- Security target ε (acceptable correctness error probability)

Ensure: Feasibility landscape $\mathcal{F}$: the set of dimensions (N, D) the device can safely run

- 1: $\mathcal{F} \leftarrow \emptyset$
- 2: **for** each circuit dimension (N, D) **do**
- 3: Sample a collection of circuits of dimension (N, D)
- 4: For each circuit, run the FK verification protocol with $T_{\text{test}} = 100$ test rounds
- 5: Let f_k = number of failed test rounds for circuit k , out of T_{test}
- 6: Compute the average test-round failure rate across all circuits:

$$\bar{p}(N, D) = \frac{1}{|\mathcal{C}|} \sum_{k \in \mathcal{C}} \frac{f_k}{T_{\text{test}}}$$

- 7: Query the verification machinery: what is the maximum tolerable noise rate $p^*(R, \varepsilon)$ such that a device with failure rate p^* can be certified within R rounds at security ε ?
- 8: **if** $\bar{p}(N, D) \leq p^*(R, \varepsilon)$ **then**
- 9: Label dimension (N, D) as *safe*: add it to $\mathcal{F}$
- 10: **end if**
- 11: **end for**
- 12: **return** $\mathcal{F}$

The key step is the comparison $\bar{p}(N, D) \leq p^*(R, \varepsilon)$: the observed failure rate is tested against the maximum noise the protocol can certify, given the available round budget R and the desired security ε . The output $\mathcal{F}$ is the feasibility landscape of the device under its observed noise.

1.3 The verification protocol

1.4 Main result (informal)

Informally, the main result is the following. Fix a threshold w/t — the maximum admissible fraction of test-round failures — and run the verification protocol on a brickwork state of dimensions (N, D) . If the observed failure rate stays

below this threshold, then the output of the computation is, with probability at least $1 - \varepsilon$, indistinguishable from the ideal output. Conversely, if the noise rate exceeds the threshold, the protocol rejects with overwhelming probability, protecting the Client from acting on a corrupted result.

More precisely, one proves that the protocol with threshold w/t is ε -close to the ideal resource, provided the circuit-level noise rate p satisfies $p < w/t - \delta$, where the security gap δ controls the exponential decay of ε with the number of test rounds. The number of rounds R required to achieve a given (ε, δ) pair constitutes the round overhead of the benchmark.

1.5 Feasibility landscape

The round overhead can be turned into a concrete benchmarking criterion. For given circuit dimensions (N, D) and a given security target ε , the verification machinery determines the minimum number of rounds R required to certify the computation. We then ask: is this R feasible in practice? Varying both ε and the allowed budget R produces a two-dimensional feasibility landscape: the set of (R, ε) pairs for which a circuit of dimensions (N, D) can be certified.

Aggregating this landscape over circuit dimensions gives a heatmap in which each cell indicates the maximum certifiable (N, D) for a given (R, ε) pair. Since the circuit dimension is two-dimensional, we reduce to a scalar size via the brick count $B(N, D)$ (defined in Eq. (1)) for display purposes, while the axes of the heatmap remain R and ε . The heatmap directly answers the question: given a round budget and a security target, how large a computation can this device run?

1.6 Simulation

Circuits are parameterised by the number of qubits N and the depth D , and are generated randomly with a BQP error rate of 0.1, yielding 100 circuits per (N, D) dimension. Each circuit is transpiled onto a pattern over a brickwork state of dimensions $N \times D$, whose entangling structure determines the circuit size.

The brickwork transpilation alternates between two layer parities: even-indexed layers contain $\lfloor N/2 \rfloor$ bricks, and odd-indexed layers contain $\lfloor (N-1)/2 \rfloor$ bricks. Over D layers — $\lceil D/2 \rceil$ even and $\lfloor D/2 \rfloor$ odd — the total brick count is

$$B(N, D) = \left\lceil \frac{D}{2} \right\rceil \left\lfloor \frac{N}{2} \right\rfloor + \left\lfloor \frac{D}{2} \right\rfloor \left\lfloor \frac{N-1}{2} \right\rfloor. \quad (1)$$

For odd N this simplifies to $B = D(N-1)/2$, since both layer types contribute $(N-1)/2$ bricks per layer. For even N it becomes $B = DN/2 - \lfloor D/2 \rfloor$. As a concrete example, for $N = 7$ and $D = 8$ (odd N), $B = 8 \times 3 = 24$ bricks.

Verification was performed using Veriphix under a depolarising noise model applied to entangling gates, at fixed noise probability p . Two representative noise regimes were studied: $p = 6 \times 10^{-4}$ and $p = 2 \times 10^{-3}$. For each circuit,

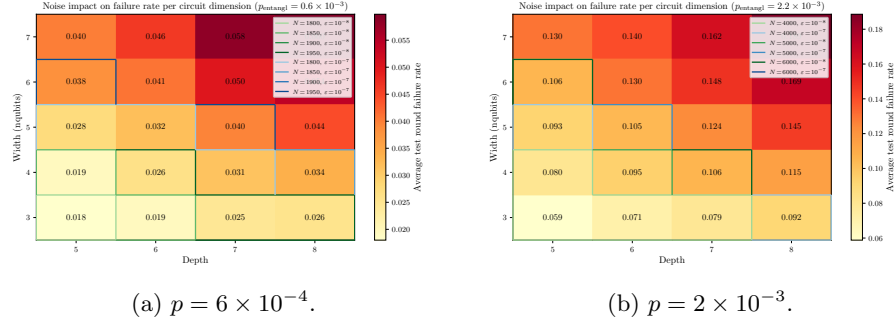


Figure 1: Discrete feasibility plot for Random Trap verification at two noise levels. Each point represents a circuit dimension (N, D) ; filled markers indicate dimensions certified as feasible (test-round failure rate below threshold), open markers indicate infeasible dimensions. The frontier (dashed) marks the boundary of certified feasibility as a function of the round budget R and target security ε .

100 test rounds were drawn using Random Traps (detection rate 1/2) and the resulting feasibility landscape was computed. The simulation was then repeated independently using the FK trap construction in place of Random Traps, enabling a direct comparison of the two trap designs at identical noise levels.

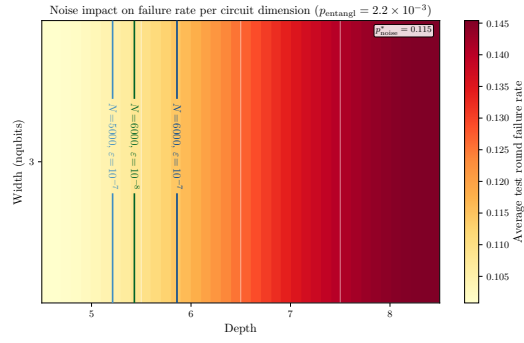


Figure 4: Feasibility heatmap for FK verification at $p = 2 \times 10^{-3}$. Layout follows Fig. 3. Direct comparison with Fig. 3(b) quantifies the impact of the trap construction on the certifiable feasibility landscape.